

Classified as hazardous in accordance with the criteria of EPA New Zealand

## Section 1 - Identification

### Product Identifier

**Product Name** Aluminum Copper square bar, alloy 2024

**Molecular Formula** Al:Cu:Mg:Mn; 93.5:4.4:1.5:0.6 wt%  
**Recommended Use** Laboratory chemicals.  
**Uses advised against** No Information available

|                                |                                                                                             |
|--------------------------------|---------------------------------------------------------------------------------------------|
| <b>Product Code</b>            | <b>42106</b>                                                                                |
| <b>Address</b>                 | Thermo Fisher Scientific New Zealand Ltd<br>244 Bush Road, Albany,<br>Auckland, New Zealand |
| <b>Emergency Tel.</b>          | <b>CHEMTREC®</b><br><b>09 980 6780 or +64 9 980 6780</b>                                    |
| <b>Telephone / Fax Numbers</b> | Tel: 09 980 6700<br>Fax: 09 980 6788                                                        |
| <b>E-mail address</b>          | <u>ANZinfo@thermofisher.com</u>                                                             |

## Section 2 - Hazard(s) Identification

### Classification under Work Safe New Zealand

Classified as hazardous in accordance with the criteria of EPA New Zealand

### GHS Classification

#### Physical hazards

Based on available data, the classification criteria are not met

#### Health hazards

|                                             |            |
|---------------------------------------------|------------|
| Acute Oral Toxicity                         | Category 3 |
| Acute Inhalation Toxicity - Dusts and Mists | Category 4 |
| Skin Sensitization                          | Category 1 |
| Germ Cell Mutagenicity                      | Category 1 |

#### Environmental hazards

|                          |            |
|--------------------------|------------|
| Chronic aquatic toxicity | Category 2 |
|--------------------------|------------|

**Label Elements** None required



**Signal Word**

**Danger**

**Hazard Statements**

H301 - Toxic if swallowed

H332 - Harmful if inhaled

H317 - May cause an allergic skin reaction

H340 - May cause genetic defects

H411 - Toxic to aquatic life with long lasting effects

**Precautionary Statements**

**Prevention**

P273 - Avoid release to the environment

**Response**

P391 - Collect spillage

**Storage**

P403 - Store in a well-ventilated place

**Disposal**

P501 - Dispose of contents/ container to an approved waste disposal plant

**Other hazards which do not result in classification**

## Section 3 - Composition and Information on Ingredients

| Component | CAS No    | Weight % |
|-----------|-----------|----------|
| Aluminum  | 7429-90-5 | 93.5     |
| Copper    | 7440-50-8 | 4.4      |
| Manganese | 7439-96-5 | 1.5      |
| Magnesium | 7439-95-4 | 0.6      |

## Section 4 - First Aid Measures

**Description of first aid measures**

**General Advice**

If symptoms persist, call a physician.

**New Zealand Emergency Tel.**

CHEMTREC®  
09 980 6780 or +64 9 980 6780

**Inhalation**

Remove to fresh air. Get medical attention immediately if symptoms occur.

**Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. Get medical attention immediately if symptoms occur.

**Ingestion**

Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.

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|                                     |                                      |
|-------------------------------------|--------------------------------------|
| Self-Protection of the First Aider  | No special precautions required.     |
| First Aid Facilities                | Eyewash, safety shower and washroom. |
| Most important symptoms and effects | None reasonably foreseeable.         |
| Notes to Physician                  | Treat symptomatically.               |

## Section 5 - Fire Fighting Measures

### **Suitable Extinguishing Media**

approved class D extinguishers. Do not use water or foam.

### **Extinguishing media which must not be used for safety reasons**

Water may be ineffective.

### **Specific Hazards Arising from the Chemical**

Thermal decomposition can lead to release of irritating gases and vapors.

### **Hazardous Combustion Products**

Metal oxides.

### **Special protective equipment and precautions for fire fighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## Section 6 - Accidental Release Measures

### Personal Precautions, Protective Equipment and Emergency Procedures

#### **Emergency procedures**

Ensure adequate ventilation. Use personal protective equipment as required. Avoid dust formation. No special precautions required.

#### **Environmental Precautions**

Do not flush into surface water or sanitary sewer system. Should not be released into the environment. Do not allow material to contaminate ground water system.

#### **Methods for Containment and Clean Up**

Sweep up and shovel into suitable containers for disposal. Avoid dust formation. Pick up and transfer to properly labelled containers.

#### **Precautions to prevent secondary hazards**

Clean contaminated objects and areas thoroughly observing environmental regulations

#### **Reference to Other Sections**

Refer to protective measures listed in Sections 8 and 13.

## Section 7 - Handling and Storage

### Precautions for Safe Handling

#### **Advice on safe handling**

Wear personal protective equipment/face protection. Ensure adequate ventilation. Avoid ingestion and inhalation. Avoid contact with skin, eyes or clothing. Avoid dust formation.

#### **Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before

re-use. Wash hands before breaks and after work.

### Conditions for Safe Storage, Including any Incompatibilities

#### Storage Conditions

Keep in a dry place. Keep away from acids.

#### Incompatible Materials

Oxidizing agent.

AS/NZS 2243.10:2004, Safety in laboratories - Storage of chemicals

## Section 8 - Exposure Controls and Personal Protection

### Control parameters

#### Exposure limits

**NZ** - Workplace Exposure Standards and Biological Exposure Indices (6th edition). New Zealand Department of Labor

**ACGIH** - Threshold Limit Values - Ceiling (TLV-C) guidelines by the American Conference of Governmental Industrial Hygienists (ACGIH) for controlling worker exposure to airborne chemical concentrations in the workplace.

**AUS** - Exposure Standards for Atmospheric Contaminants in the Occupational Environment - Guidance Note on the Interpretation of Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:3008(1995)]

Adopted National Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:1003(1995)] updated in August, 2005. Safe Work Australia

**UK** - EH40/2005 Work Exposure Limits, Fourth edition. Published 2020.

| Component | New Zealand WEL                                           | Australia                                              | ACGIH TLV                                                 | The United Kingdom                                                                                                                               |
|-----------|-----------------------------------------------------------|--------------------------------------------------------|-----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Aluminum  | TWA: 10 mg/m <sup>3</sup>                                 | TWA: 10 mg/m <sup>3</sup><br>TWA: 5 mg/m <sup>3</sup>  | TWA: 1 mg/m <sup>3</sup>                                  | STEL: 30 mg/m <sup>3</sup> 15 min<br>STEL: 12 mg/m <sup>3</sup> 15 min<br>TWA: 10 mg/m <sup>3</sup> 8 hr<br>TWA: 4 mg/m <sup>3</sup> 8 hr        |
| Copper    | TWA: 0.01 mg/m <sup>3</sup>                               | TWA: 1 mg/m <sup>3</sup><br>TWA: 0.2 mg/m <sup>3</sup> | TWA: 0.2 mg/m <sup>3</sup>                                | STEL: 0.6 mg/m <sup>3</sup> 15 min<br>STEL: 2 mg/m <sup>3</sup> 15 min<br>TWA: 1 mg/m <sup>3</sup> 8 hr<br>TWA: 0.2 mg/m <sup>3</sup> 8 hr       |
| Manganese | TWA: 0.2 mg/m <sup>3</sup><br>TWA: 0.02 mg/m <sup>3</sup> | STEL: 3 mg/m <sup>3</sup><br>TWA: 1 mg/m <sup>3</sup>  | TWA: 0.02 mg/m <sup>3</sup><br>TWA: 0.1 mg/m <sup>3</sup> | STEL: 0.6 mg/m <sup>3</sup> 15 min<br>STEL: 0.15 mg/m <sup>3</sup> 15 min<br>TWA: 0.2 mg/m <sup>3</sup> 8 hr<br>TWA: 0.05 mg/m <sup>3</sup> 8 hr |

#### Biological limit values

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

### Appropriate engineering controls

#### Engineering Measures

None under normal use conditions.

### Individual protection measures, such as personal protective equipment

#### Eye Protection

Wear safety glasses with side shields (or goggles) (Australian/New Zealand Standard AS/NZS 1337 - Eye protectors for Industrial applications)

#### Hand Protection

No special protective equipment required

| Glove material                                 | Breakthrough time                 | Glove thickness | AUS/NZ Standard | Glove comments        |
|------------------------------------------------|-----------------------------------|-----------------|-----------------|-----------------------|
| Natural rubber, Nitrile rubber, Neoprene, PVC. | See manufacturers recommendations | -               | AS/NZS 2161     | (minimum requirement) |

#### Skin and body protection

Long sleeved clothing

|                                        |                                                                                                                                                                                                                                                                                                                             |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Respiratory Protection</b>          | Use an AS/NZS 1716 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained in line with AS/NZS 1715 on the use and maintenance of respiratory protective devices |
| <b>Recommended Filter type:</b>        | Particle filter (or AUS/NZ equivalent)                                                                                                                                                                                                                                                                                      |
| <b>Hygiene Measures</b>                | Handle in accordance with good industrial hygiene and safety practice.                                                                                                                                                                                                                                                      |
| <b>Environmental exposure controls</b> | Prevent product from entering drains. Do not allow material to contaminate ground water system. Local authorities should be advised if significant spillages cannot be contained.                                                                                                                                           |

## Section 9 - Physical and Chemical Properties

### Information on basic physical and chemical properties

|                                                |                                   |                                          |
|------------------------------------------------|-----------------------------------|------------------------------------------|
| <b>Physical State</b>                          | Solid Bar                         |                                          |
| <b>Appearance</b>                              | Silver                            |                                          |
| <b>Odor</b>                                    | Odorless                          |                                          |
| <b>Odor Threshold</b>                          | No data available                 |                                          |
| <b>pH</b>                                      | No information available          |                                          |
| <b>Melting Point/Range</b>                     | No data available                 |                                          |
| <b>Softening Point</b>                         | No data available                 |                                          |
| <b>Boiling Point/Range</b>                     | No information available          |                                          |
| <b>Flammability (liquid)</b>                   | Not applicable                    | Solid                                    |
| <b>Flammability (solid,gas)</b>                | No information available          |                                          |
| <b>Explosion Limits</b>                        | No data available                 |                                          |
| <b>Flash Point</b>                             | No information available          | <b>Method -</b> No information available |
| <b>Autoignition Temperature</b>                | No data available                 |                                          |
| <b>Decomposition Temperature</b>               | No data available                 |                                          |
| <b>Viscosity</b>                               | Not applicable                    | Solid                                    |
| <b>Water Solubility</b>                        | Insoluble in water                |                                          |
| <b>Solubility in other solvents</b>            | No information available          |                                          |
| <b>Partition Coefficient (n-octanol/water)</b> |                                   |                                          |
| <b>Vapor Pressure</b>                          | 23 hPa @ 20 °C                    |                                          |
| <b>Density / Specific Gravity</b>              | No data available                 |                                          |
| <b>Bulk Density</b>                            | No data available                 |                                          |
| <b>Vapor Density</b>                           | Not applicable                    | Solid                                    |
| <b>Particle characteristics</b>                | No data available                 |                                          |
| <b><u>Other information</u></b>                |                                   |                                          |
| <b>Molecular Formula</b>                       | Al:Cu:Mg:Mn; 93.5:4.4:1.5:0.6 wt% |                                          |
| <b>Evaporation Rate</b>                        | Not applicable - Solid            |                                          |

## Section 10 - Stability and Reactivity

|                                         |                                            |
|-----------------------------------------|--------------------------------------------|
| <b>Reactivity</b>                       | None known, based on information available |
| <b>Stability</b>                        | Stable under normal conditions.            |
| <b>Sensitivity to Mechanical Impact</b> | No information available                   |
| <b>Sensitivity to Static Discharge</b>  | No information available                   |
| <b>Hazardous Polymerization</b>         | No information available.                  |
| <b>Hazardous Reactions</b>              | None under normal processing.              |

**Conditions to Avoid** Heat, flames and sparks.

**Incompatible Materials** Oxidizing agent.

**Hazardous Decomposition Products** Metal oxides.

## **Section 11 - Toxicological Information**

### **Acute Effects**

### **Information on likely routes of exposure**

#### **Product Information**

|                   |                                                |
|-------------------|------------------------------------------------|
| <b>Inhalation</b> | Not an expected route of exposure.             |
| <b>Eyes</b>       | Not an expected route of exposure.             |
| <b>Skin</b>       | No known effect based on information supplied. |
| <b>Ingestion</b>  | Not an expected route of exposure.             |

### **Numerical measures of toxicity**

#### **(a) acute toxicity;**

|                   |                                                                  |
|-------------------|------------------------------------------------------------------|
| <b>Oral</b>       | Based on available data, the classification criteria are not met |
| <b>Dermal</b>     | No data available                                                |
| <b>Inhalation</b> | No data available                                                |

### **Toxicology data for the components**

| Component | LD50 Oral                | LD50 Dermal | LC50 Inhalation               |
|-----------|--------------------------|-------------|-------------------------------|
| Aluminum  |                          |             | LC50 > 0.888 mg/L ( Rat ) 4 h |
| Copper    |                          |             | LC50 > 5.11 mg/L ( Rat ) 4 h  |
| Manganese | LD50 = 9 g/kg ( Rat )    |             | LC50 > 5.14 mg/L ( Rat ) 4 h  |
| Magnesium | LD50 = 230 mg/kg ( Rat ) |             |                               |

**(b) skin corrosion/irritation;** No data available

**(c) serious eye damage/irritation;** No data available

#### **(d) respiratory or skin sensitization;**

|                    |                   |
|--------------------|-------------------|
| <b>Respiratory</b> | No data available |
| <b>Skin</b>        | No data available |

**(e) germ cell mutagenicity;** No data available

**(f) carcinogenicity;** No data available

There are no known carcinogenic chemicals in this product

**(g) reproductive toxicity;** No data available

**(h) STOT-single exposure;** No data available

(i) STOT-repeated exposure; No data available

Target Organs No information available.

(j) aspiration hazard; Not applicable  
Solid

Symptoms / effects, both acute and delayed  
No information available.

## Section 12 - Ecological Information

### Ecotoxicity

**Aquatic ecotoxicity** Contains a substance which is: Very toxic to aquatic organisms. The product contains following substances which are hazardous for the environment. May cause long-term adverse effects in the environment. Do not allow material to contaminate ground water system.

| Component | Freshwater Fish                                                                      | Water Flea                                       | Freshwater Algae                                                 | Microtox |
|-----------|--------------------------------------------------------------------------------------|--------------------------------------------------|------------------------------------------------------------------|----------|
| Copper    | Onchorhynchys mykiss:<br>LC50=0.15 mg/L 96h<br>Cuprinus carpio:<br>LC50=0.8 mg/L 96h | EC50: = 0.03 mg/L, 48h<br>Static (Daphnia magna) | 0.0426-0.0535 mg/L<br>EC50 72 h<br>0.031-0.054 mg/L EC50<br>96 h |          |
| Manganese | LC50: > 3.6 mg/L, 96h<br>semi-static<br>(Oncorhynchus mykiss)                        |                                                  |                                                                  |          |

**Terrestrial ecotoxicity** There is no data for this product

**Persistence and Degradability** Product contains heavy metals. Discharge into the environment must be avoided. Special pre-treatment is necessary

**Persistence** Insoluble in water, May persist.

**Degradability** Not relevant for inorganic substances.  
**Degradation in sewage treatment plant** Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

**Bioaccumulative Potential** Product has a high potential to bioconcentrate May have some potential to bioaccumulate

**Mobility** Spillage unlikely to penetrate soil. Is not likely mobile in the environment due its low water solubility.

### Other adverse effects

**Endocrine Disruptor Information** This product does not contain any known or suspected endocrine disruptors  
**Persistent Organic Pollutant** This product does not contain any known or suspected substance  
**Ozone Depletion Potential** This product does not contain any known or suspected substance

## Section 13 - Disposal Considerations

### Waste treatment methods

|                                            |                                                                                                                                                                                                                                                                                                                      |
|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Waste from Residues/Unused Products</b> | Do not allow into drains or watercourses or dispose of where ground or surface waters may be affected. Wastes, including emptied containers, are controlled wastes and should be disposed of in accordance with all federal, E.P.A., state and local regulations. Assure conformity with all applicable regulations. |
| <b>Contaminated Packaging</b>              | Dispose of this container to hazardous or special waste collection point.                                                                                                                                                                                                                                            |
| <b>Other Information</b>                   | Disposal agencies or waste contractors must comply with the New Zealand Hazardous Substances (Disposal) Regulations . Do not flush to sewer.                                                                                                                                                                         |

## Section 14 - Transport Information

| Component                      | Hazchem Code   |
|--------------------------------|----------------|
| Magnesium<br>7439-95-4 ( 0.6 ) | 1Z<br>4Y<br>4W |

**NZS 5433:2020** Not regulated

**IATA** Not regulated

**IMDG/IMO** Not regulated

| Component                   | IMDG Marine Pollutant                                                                   |
|-----------------------------|-----------------------------------------------------------------------------------------|
| Copper<br>7440-50-8 ( 4.4 ) | IMDG regulated marine pollutant (Listed in the index, listed under Copper metal powder) |

**Environmental hazards** No hazards identified

**Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code** Not applicable, packaged goods

**Special Precautions** No special precautions required. Please refer to the applicable dangerous goods regulations for additional information.

**Additional information** None known

## Section 15 - Regulatory Information

### Safety, health and environmental regulations/legislation specific for the substance or mixture

#### National Regulations

Any applicable tolerable exposure limits and environmental exposure limits according to the EPA Controls for Hazardous Substances are listed below

| Component | Tolerable Exposure Limit (TEL) Air | Tolerable Exposure Limit (TEL) Water | Tolerable Exposure Limit (TEL) Surface | Environmental Exposure Limits (EEL)                    |
|-----------|------------------------------------|--------------------------------------|----------------------------------------|--------------------------------------------------------|
| Copper    |                                    |                                      |                                        | 0.0014 mg/L (Freshwater)<br>0.0013 mg/L (Marine water) |

#### Certified handlers, tracking and controlled substance license requirements

Certified handlers are required for some substances. This includes substances requiring a controlled substance license, and most explosives, vertebrates toxic agents, and certain fumigants. Acutely toxic substances which are a Category 1 or 2, such as pesticides also require Certified handlers. Please check the Health and Safety at Work Act 2015 for further information. Tracking is required for some highly hazardous substances. These substances need to be under the control of an appropriately trained person or appropriately secured. Please check the Health and Safety at Work Act 2015 for further information.

#### Prohibition or notification/licensing requirements



Shown below are details of specific prohibition/notifications or licencing requirements when they apply.

### International Regulations

**Ozone Depletion Potential** This product does not contain any known or suspected substance

**Persistent Organic Pollutant** This product does not contain any known or suspected substance

**Rotterdam Convention (PIC)** Not applicable

| Component | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | IMDG Marine Pollutant                                                                   |
|-----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| Copper    |                                                                                           |                                                                                          | IMDG regulated marine pollutant (Listed in the index, listed under Copper metal powder) |

### Authorisation/Restrictions according to EU REACH

| Component | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|-----------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Aluminum  | -                                                                   | Use restricted. See item 75. (see link for restriction details)               | -                                                                                                     |
| Copper    | -                                                                   | Use restricted. See item 75. (see link for restriction details)               | -                                                                                                     |

<https://echa.europa.eu/substances-restricted-under-reach>

### International Inventories

New Zealand (NZIoC), Australia (AICS), Europe (EINECS/ELINCS/NLP), Korea (KECL), China (IECSC), Taiwan (TCSI), Japan (ENCS), Japan (ISHL), Canada (DSL/NDL), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

| Component | CAS No    | NZIoC | AICS | EINECS | ELINCS | NLP | KECL     | IECSC | TCSI |
|-----------|-----------|-------|------|--------|--------|-----|----------|-------|------|
| Aluminum  | 7429-90-5 | X     | X    | -      | -      | -   | KE-00881 | X     | X    |
| Copper    | 7440-50-8 | X     | X    | -      | -      | -   | KE-08896 | X     | X    |
| Manganese | 7439-96-5 | X     | X    | -      | -      | -   | KE-22999 | X     | X    |
| Magnesium | 7439-95-4 | X     | X    | -      | -      | -   | KE-22673 | X     | X    |

| Component | CAS No    | TSCA | TSCA Inventory notification - Active-Inactive | DSL | NDL | PICCS | ISHL | ENCS |
|-----------|-----------|------|-----------------------------------------------|-----|-----|-------|------|------|
| Aluminum  | 7429-90-5 | X    | ACTIVE                                        | X   | -   | X     | -    | X    |
| Copper    | 7440-50-8 | X    | ACTIVE                                        | X   | -   | X     | -    | X    |
| Manganese | 7439-96-5 | X    | ACTIVE                                        | X   | -   | X     | -    | X    |
| Magnesium | 7439-95-4 | X    | ACTIVE                                        | X   | -   | X     | -    | X    |

**Legend:** X - Listed '-' - Not Listed

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

## Section 16 - Other Information

This safety data sheet complies with the requirements of the EPA Hazardous Substances (Hazard Classification) Notice 2020 and WorkSafe New Zealand Regulations

### Legend

**NZIoC** - New Zealand Inventory of Chemicals

**AICS** - Australian Inventory of Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b)

#### Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**IECSC** - Chinese Inventory of Existing Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**NZS 5433:2020** - Transport of Dangerous Goods on Land

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**WEL** - Workplace Exposure Limit

**DNEL** - Derived No Effect Level

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**VOC** - (Volatile Organic Compound)

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**ENCS** - Japanese Existing and New Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**CAS** - Chemical Abstracts Service

**ACGIH** - American Conference of Governmental Industrial Hygienists

**PNEC** - Predicted No Effect Concentration

**OECD** - Organisation for Economic Co-operation and Development

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**ADG** - Australian Code for the Transport of Dangerous Goods by Road and Rail

**LC50** - Lethal Concentration 50%

**ATE** - Acute Toxicity Estimate

**RPE** - Respiratory Protective Equipment

**NOEC** - No Observed Effect Concentration

**BCF** - Bioconcentration factor

**PBT** - Persistent, Bioaccumulative, Toxic

#### Key literature references and sources for data

HSNO classifications provided in the New Zealand Chemical Classification Information Database (CCID).

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

EPA Guide to classifying hazardous substances in New Zealand

EPA - Assigning a product to an existing HSNO approval guide

#### Classification and procedure used to derive the classification for mixtures according to Regulation (EC) 1272/2008 [CLP]:

**Physical hazards** On basis of test data

**Health Hazards** Calculation method

**Environmental hazards** Calculation method

#### Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

**Revision Date** 22-Mar-2023

**Revision Summary** Not applicable

#### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**